Decision Brief: NEON Site Compatibility

ExSOIL NSF Prototype · June 2026 · Steven Wangen

Open for Discussion

Problem

The NEON site workflow is non-functional in the rebuilt container.

CESM 2.2.2's CLM component predates the NEON tower site workflow. The per-site configuration directories ("usermods") that `run_neon_v2.py` depends on do not exist in this CLM release. Running any NEON site command fails immediately:

```
$ run_neon_v2 --neon-sites ABBY --output-root ~/output --overwrite

run_neon_v2: error: argument --neon-sites: invalid choice: 'ABBY' (choose from 'all')
```

Why this happens

The script discovers available sites by scanning a directory in the CLM source tree:

```
valid_neon_sites = sorted([
    v.split("/")[-1]
    for v in glob.glob(
        os.path.join(cesmroot, "cime_config", "usermods_dirs", "NEON", "[!d]*")
    )
])
```

In the original `escomp/cesm-lab-neon` container (built ~2022), this directory contained 81 NEON site folders (ABBY, BART, KONZ, etc.). In CESM 2.2.2's standard CLM checkout, **this directory does not exist**. The NEON usermods were added to CTSM after the 2.2.x release series.

What still works

Capability	Status	Notes
General CESM case creation	**Working**	I2000Clm50Sp, B1850, etc. all verified
CESM Fortran compilation	**Working**	case.build produces cesm.exe in 78s on arm64
Python scientific stack	**Working**	All 28 tutorial packages, 63 smoke tests pass
JupyterLab	**Working**	Starts, serves notebooks, Panel extensions loaded
NEON site runs (run_neon_v2)	**Broken**	No valid site codes; usermods directory missing
Design_Hub / Modeling_Hub notebooks	**Broken**	Depend on NEON site runs

Options

Option A: Graft NEON usermods from newer CTSM

Low effort Moderate risk

Copy the `usermods_dirs/NEON/` directory from a recent CTSM release (e.g., ctsm5.2) into the CESM 2.2.2 CLM tree during the Docker build. This is likely how the original ESCOMP image was assembled.

Pros:

- Minimal disruption to the rest of the CESM source tree
- NEON usermods are mostly configuration files (XML, shell scripts), not compiled code
- A few `COPY` lines in the Dockerfile

Cons:

- Usermods may reference namelist options or surface datasets that don't exist in CESM 2.2.2's CLM
- Compatibility not guaranteed; could cause subtle configuration errors
- Creates ongoing maintenance burden tracking which CTSM version usermods came from

Timeline: Hours to implement, days to validate.

Option B: Pin CLM component to newer CTSM tag

Medium effort | Moderate risk

Override just the CLM/CTSM component in the CESM checkout to use a newer tag (e.g., `ctsm5.2.005`) while keeping the rest of CESM at 2.2.2. CESM's `Externals.cfg` supports per-component tag overrides.

Pros:

- Gets the full NEON workflow as designed: usermods, surface datasets, code fixes
- Officially supported CLM version with community testing
- Forward-compatible with future NEON site additions

Cons:

- Newer CTSM may have API incompatibilities with CESM 2.2.2's CIME or coupler
- Could break the case.build compilation (89/89 tests currently pass)
- Mixing component versions is officially unsupported by ESCOMP

Timeline: Days. Requires rebuild, recompile, full re-test.

Option C: Bundle NEON configs in this repository

Medium-high effort | Low risk

Create a `neon-sites/` directory in this repository with per-site usermods (shell_commands, user_nl_clm, etc.) and copy them into the CLM tree during the Docker build. Maintain independently of any CTSM release.

Pros:

- Full control over site configurations; no external version dependency
- Can limit to the specific sites the project uses (not all 81)
- Easy to add custom sites or project-specific modifications

Cons:

- Significant upfront effort to create or port usermods for each site
- Must manually maintain surface dataset references and domain file paths
- Diverges from the community CTSM NEON workflow

Timeline: Days to a week, depending on number of sites needed.

Option D: Upgrade to CESM 3.x / CTSM 6.x

High effort | High risk

Move to a current CESM release that natively includes the NEON workflow. Resolves this issue at its root and also fixes the Python 3.11 pin, the PIO2 filter patches, and other 2.2.x limitations.

Pros:

- Resolves NEON, Python version, PIO2, and other 2.2.x issues in one move
- Aligns with the community's current supported version
- Future-proofs the container for years

Cons:

- Major version upgrade with potentially breaking changes across all components
- `run_neon_v2.py` was written against 2.2.x APIs; likely needs significant rewrite
- All 89 container tests need re-validation
- Machine configurations and compiler flags may need rework

Timeline: Weeks. Essentially a second rebuild effort.

Comparison

	A: Graft usermods	B: Newer CLM tag	C: Bundle in repo	D: CESM 3.x
Effort	**Low**	**Medium**	**Medium-high**	**High**
NEON sites work	**Likely**	**Yes**	**Yes**	**Yes**
Existing tests pass	**Yes**	**Unknown**	**Yes**	**Must re-validate**
Fixes Python 3.11 pin	**No**	**No**	**No**	**Yes**
Fixes PIO2 patches	**No**	**No**	**No**	**Yes**
Community-maintained	**Partial**	**Yes**	**No**	**Yes**
Risk to current build	**Low**	**Moderate**	**Low**	**High**

Questions for Discussion

Q1 **Which NEON sites does the project actively use?** If it's 3-5 sites, Option C is attractive. If many sites or all 81, Options A or B scale better.

Q2 **How tightly coupled is `run_neon_v2.py` to CESM 2.2.x internals?** If it mostly uses CIME's public API (create_newcase, xmlchange), a component upgrade (Option B) is safer. If it patches internal CIME files, it's riskier.

Q3 **Is there appetite for a CESM 3.x migration?** If the project plans to upgrade eventually, doing it now (Option D) avoids a throwaway intermediate step. If 2.2.x is the target for the foreseeable future, a lighter option makes more sense.

Q4 **Did the original ESCOMP container use a custom CTSM branch?** Identifying exactly which CLM/CTSM the `escomp/cesm-lab-neon` image used would clarify the compatibility boundary.

Context

This issue was discovered during the multi-platform container rebuild (see rebuild report). The rebuild successfully modernized the base image, achieved native arm64 support with 89/89 tests passing, and resolved 13 compatibility issues between CESM 2.2.2 and modern toolchains. The NEON site issue is separate from the multi-arch work and predates it: it is a consequence of moving from the ESCOMP-customized container to a standard CESM 2.2.2 checkout.